

# CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

**PRE-MEDICAL : ENTHUSIAST COURSE [ PHASE-I ]**

**TEST # 05**

**DATE : 24-11-2019**

**REVISION TEST**

**Test Pattern : NEET**

## INSTRUCTIONS

**Do not open this Test Booklet until you are asked to do so**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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**REVISION TEST # 05**

Topics

DATE : 24 - 11 - 2019

**PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]**

PHYSICS : Electrostatics

CHEMISTRY : Gaseous State, Periodic Table

ZOOLOGY : Nervous System &amp; Sense Organs

SPACE FOR ROUGH WORK

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

**BEWARE OF NEGATIVE MARKING**

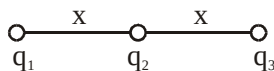
1. Two identical conducting spheres M and N has charges  $q_m$  and  $q_n$  respectively. A third identical neutral sphere P is brought in contact with M and then separated. Now sphere P is brought in contact with N then final charge on sphere P is

(1)  $\frac{q_m + 2q_n}{4}$  (2)  $\frac{q_m + q_n}{4}$   
(3)  $q_m + \frac{q_n}{2}$  (4)  $\frac{q_m + 2q_n}{2}$

2. In a uniform electric field, a cube of side 1 cm is placed. The total energy stored in the cube is  $8.85\mu\text{J}$ . The electric field is parallel to four of the faces of the cube. The electric flux through any one of the remaining two faces is.

(1)  $\frac{1}{5\sqrt{2}} \text{ V/m}$  (2)  $100\sqrt{2} \text{ V/m}$   
(3)  $5\sqrt{2} \text{ V/m}$  (4)  $10\sqrt{2} \text{ V/m}$

3. Three charged particles are placed on a straight line as shown in Fig.  $q_1$  and  $q_2$  are fixed, but  $q_3$  can be moved. Under the action of the forces from  $q_1$  and  $q_2$ ,  $q_3$  is in equilibrium. What is the relation between  $q_1$  and  $q_2$ ?



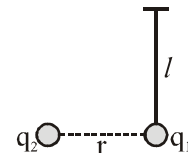
(1)  $q_1 = 4q_2$  (2)  $q_1 = -q_2$   
(3)  $q_1 = -4q_2$  (4)  $q_1 = q_2$

4. Two particles A and B (B is right of A) having charges  $8 \times 10^{-6} \text{ C}$  and  $-2 \times 10^{-6} \text{ C}$ , respectively, are held fixed with separation of 20 cm. Where should a third charged particle be placed so that it does not experience a net electric force :-

- (1) 5 cm right of B  
(2) 5 cm left of A  
(3) 20 cm left of A  
(4) 20 cm right of B

5. An isolated charge  $q_1$  of mass  $m$  is suspended freely by a thread of length  $l$ . Another charge  $q_2$  is brought near it ( $r \gg l$ ). When  $q_1$  is in equilibrium, tension in thread will be :-

- (1)  $mg$   
(2)  $>mg$   
(3)  $<mg$   
(4) none of these

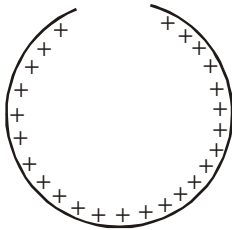


6. An oil drop, carrying six electronic charges and having a mass of  $1.6 \times 10^{-12} \text{ g}$ , falls with some terminal velocity in a medium. What magnitude of vertical electric field is required to make the drop move upward with the same speed as it was formerly moving downward with? Ignore buoyancy :-

- (1)  $10^5 \text{ NC}^{-1}$   
(2)  $10^4 \text{ NC}^{-1}$   
(3)  $3.3 \times 10^4 \text{ NC}^{-1}$   
(4)  $3.3 \times 10^5 \text{ NC}^{-1}$

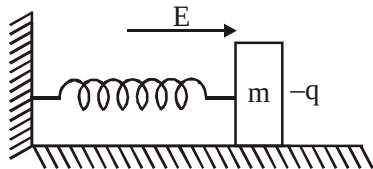
SPACE FOR ROUGH WORK

7. A ring of charge with radius 0.5 m has  $0.002\pi$  m gap. If the ring carries a charge of +1 C, the electric field at the centre is :-



- (1)  $7.5 \times 10^7 \text{ NC}^{-1}$  (2)  $7.2 \times 10^7 \text{ NC}^{-1}$   
(3)  $6.2 \times 10^7 \text{ NC}^{-1}$  (4)  $6.5 \times 10^7 \text{ NC}^{-1}$

8. A block of mass  $m$  containing a net negative charge  $-q$  is placed on a frictionless horizontal table and is connected to a wall through an unstretched spring of spring constant  $k$ . If the horizontal electric field  $E$  parallel to the spring is switched on, then the maximum compression of the spring is :-



- (1)  $\sqrt{qE/k}$  (2)  $2qE/k$   
(3)  $qE/k$  (4) zero

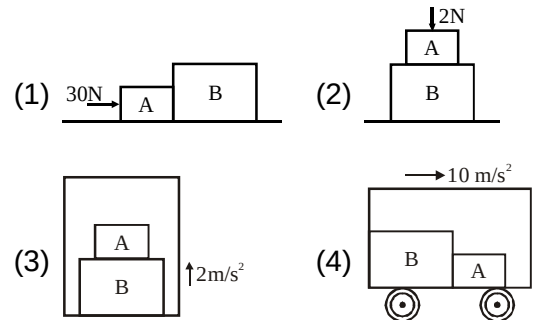
9. Five point charges, each of value  $+q$ , are placed on five vertices of a regular hexagon of side  $L$ . The magnitude of the force on a point charge of value  $-q$  coulomb placed at the center of the hexagon is

- (1)  $\frac{1}{\pi\epsilon_0} \left(\frac{q}{L}\right)^2$  (2)  $\frac{2}{\pi\epsilon_0} \left(\frac{q}{L}\right)^2$   
(3)  $\frac{1}{2\pi\epsilon_0} \left(\frac{q}{L}\right)^2$  (4)  $\frac{1}{4\pi\epsilon_0} \left(\frac{q}{L}\right)^2$

10. It is required to hold equal charges  $q$  in equilibrium at the corners of a square. What charge when placed at the centre of the square will do this ?

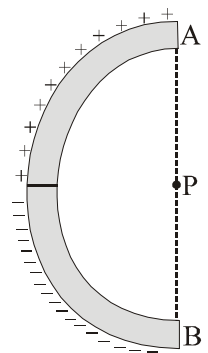
- (1)  $-\frac{q}{2}(1+2\sqrt{2})$  (2)  $\frac{q}{2}(1+2\sqrt{2})$   
(3)  $\frac{q}{4}(1+2\sqrt{2})$  (4)  $-\frac{q}{4}(1+2\sqrt{2})$

11. In which of the following cases is the contact force between A and B is maximum. If mass of each block is 1 kg :-



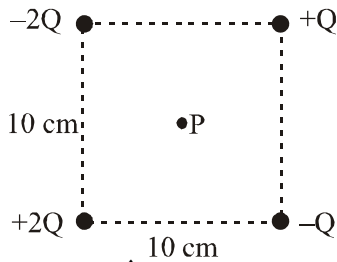
12. A thin glass rod is bent into a semicircle of radius  $r$ . A charge  $+Q$  is uniformly distributed along the upper half, and a charge  $-Q$  is uniformly distributed along the lower half, as shown in Fig. The electric field  $E$  at  $P$ , the center of the semicircle, is :-

- (1)  $\frac{Q}{\pi^2\epsilon_0 r^2}$   
(2)  $\frac{2Q}{\pi^2\epsilon_0 r^2}$   
(3)  $\frac{4Q}{\pi^2\epsilon_0 r^2}$   
(4)  $\frac{Q}{4\pi^2\epsilon_0 r^2}$



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13. Four electrical charges are arranged on the corners of a 10 cm square as shown. What would be the direction of the resulting electric field at the center point P?



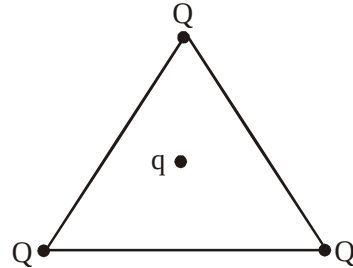
(1)  $\rightarrow$  (2)  $\uparrow$  (3)  $\leftarrow$  (4)  $\downarrow$

14. Two pith balls each with mass  $m$  are suspended from insulating threads. When the pith balls are given equal positive charge  $Q$ , they hang in equilibrium as shown. We now increase the charge on the left pith ball from  $Q$  to  $2Q$  while leaving its mass essentially unchanged. Which of the following diagrams best represents the new equilibrium configuration?



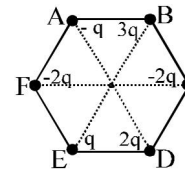
- (1) (2) (3) (4)

15. Three charges (each  $Q$ ) are placed at the three corners of an equilateral triangle. A fourth charge  $q$  is placed at the centre of the triangle. The ratio  $|q/Q|$  so as to make the system in equilibrium is :-



- (1) 1.3 (2)  $1:\sqrt{3}$   
(3)  $\sqrt{3}:1$  (4)  $2:\sqrt{3}$

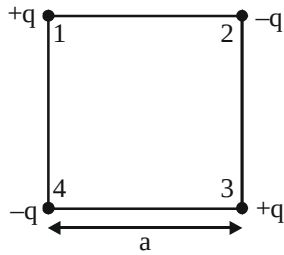
16. Six charges are placed at the corner of a regular hexagon as shown. If an electron is placed at its centre O, force on it will be:



- (1) Zero (2) Along OF  
(3) Along OC (4) None of these
17. Two equal negative charges each  $-q$ , are placed at points  $(0, a)$  and  $(0, -a)$  on  $y$ -axis. A positive charge  $q$  is released from point  $(2a, 0)$ . This charge will be :-
- (1) execute SHM about the origin  
(2) oscillate but not execute SHM  
(3) move towards origin and will become stationary  
(4) execute SHM along  $y$ -axis

SPACE FOR ROUGH WORK

18. The work done required to put the four charges together at the corners of a square of side  $a$ , as shown in the figure is :-

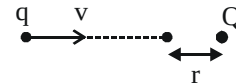


- (1)  $\frac{1}{4\pi\epsilon_0} \frac{q^2}{a}$  (2)  $-\frac{2.6}{4\pi\epsilon_0} \frac{q^2}{a}$
- (3)  $+\frac{2.6}{4\pi\epsilon_0} \frac{q^2}{a}$  (4) none of these
19. The net electric flux  $\phi$  through a hemisphere surface of radius  $R$ , placed in a uniform electric field of intensity  $E$  which is parallel to the axis of its circular plane is :-
- (1)  $2\pi RE$  (2)  $2\pi R^2 E$
- (3) Zero (4)  $(4/3)\pi R^3 E$
20. Mark the correct statement about the electric lines of force :-
- (1) The number of lines of force leaving a +ve charge or entering a -ve charge is proportional to the magnitude of charge
- (2) The path of a charged particle in an electric field is not always along the field lines
- (3) The direction of electric lines of force is along the normal to the conducting surfaces
- (4) All of the above

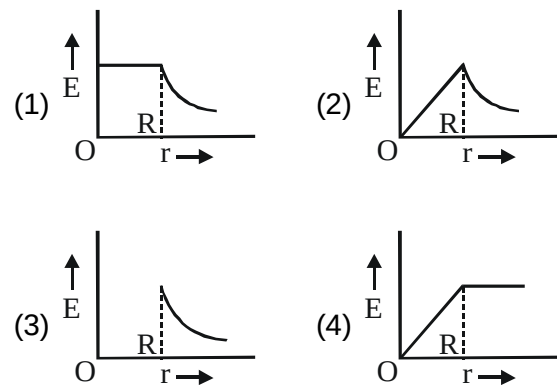
21. A circular ring of radius  $R$  which has uniformly distributed charge  $Q$ . Find the flux crossing through a sphere of radius  $R$  having its centre on the periphery of the ring :-

- (1) zero (2)  $\frac{Q}{4\epsilon_0}$
- (3)  $\frac{Q}{\epsilon_0}$  (4)  $\frac{Q}{3\epsilon_0}$

22. A charged particle  $q$  is shot from far distance towards another charged particle  $Q$  which is fixed, with a speed  $v$ . It approaches  $Q$  upto a closest distance  $r$  and then returns. If  $q$  were given a speed  $2v$ , the closest distance of approach would be :-

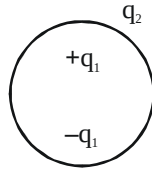


- (1)  $r$  (2)  $2r$  (3)  $r/2$  (4)  $r/4$
23. The electric field due to a uniformly charged non conducting sphere of radius  $R$  as a function of the distance from its centre is represented graphically by :-

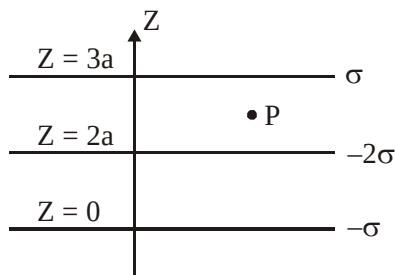


SPACE FOR ROUGH WORK

24. Consider the charge configuration and a spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to:-



- (1)  $q_2$   
(2) only the positive charges  
(3) all the charges  
(4)  $+q_1$  and  $-q_1$
25. Three infinitely charged non conducting sheets are kept parallel to XY-plane having charge densities as shown in figure. Find out the net electric field at point P :-



- (1)  $\frac{2\sigma}{\epsilon_0} \hat{k}$   
(2)  $-\frac{2\sigma}{\epsilon_0} \hat{k}$   
(3)  $-\frac{4\sigma}{\epsilon_0} \hat{k}$   
(4)  $\frac{4\sigma}{\epsilon_0} \hat{k}$

26. Positive and negative point charges of equal magnitude are kept at  $\left(0, 0, \frac{a}{2}\right)$  and

$\left(0, 0, \frac{-a}{2}\right)$  respectively. The work done by the electric field when another positive point charge is moved from  $(-a, 0, 0)$  to  $(0, a, 0)$  is :-

- (1) positive  
(2) negative  
(3) zero  
(4) dependent on the path connecting the initial and final positions
27. Two infinitely long parallel wires having linear charge densities  $\lambda_1$  and  $\lambda_2$  respectively which are placed at a perpendicular distance of R metres. The force per unit length on either wire will

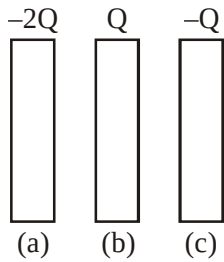
be  $\left(K = \frac{1}{4\pi\epsilon_0}\right)$  :-

- (1)  $K \frac{2\lambda_1\lambda_2}{R^2}$  (2)  $K \frac{2\lambda_1\lambda_2}{R}$   
(3)  $K \frac{\lambda_1\lambda_2}{R^2}$  (4)  $K \frac{\lambda_1\lambda_2}{R}$
28. An electron (of charge  $-e$ ) revolves around a long wire with uniform charge density  $\lambda$  in a circular path of radius r. Its kinetic energy is given by :-

- (1)  $\frac{\lambda e}{2\pi\epsilon_0 r}$  (2)  $\frac{\lambda e}{4\pi\epsilon_0 r}$   
(3)  $\frac{\lambda e}{2\pi\epsilon_0}$  (4)  $\frac{\lambda e}{4\pi\epsilon_0}$

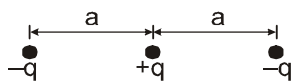
SPACE FOR ROUGH WORK

29. Three identical conducting plates are shown in figure. Find out the charge on the right face of the plate b.



- (1)  $-Q$  (2)  $\frac{Q}{2}$  (3)  $Q$  (4) Zero

30. The three point charges shown in the figure lie along a straight line. The energy needed to exchange the position of the central +ve charge with one of the negative charges is :-

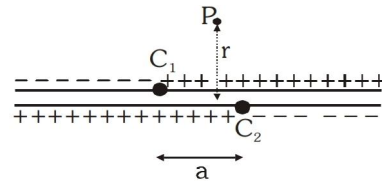


- (1)  $\frac{q^2}{8\pi\epsilon_0 a}$  (2)  $\frac{3q^2}{8\pi\epsilon_0 a}$   
(3)  $\frac{q^2}{4\pi\epsilon_0 a}$  (4)  $\frac{-q^2}{8\pi\epsilon_0 a}$

31. A point charge  $q = 100 \mu\text{C}$  is located in the x-y plane at the point with position vector  $= \vec{r}_0 = 2\hat{i} + 3\hat{j} + \hat{k}$ . What is the electric field vector (in kV/m) at the point with position vector  $= \vec{r} = 8\hat{i} - 5\hat{j} + \hat{k}$ ?

- (1)  $4.2\hat{i} - 5.6\hat{j}$  (2)  $6\hat{i} - 8\hat{j}$   
(3)  $54\hat{i} - 72\hat{j}$  (4)  $5.4\hat{i} - 7.2\hat{j}$

32. Two long thin rods having charge density  $\lambda_0 x$  and  $-\lambda_0 x$  (where  $x$  is the distance from their respective centres  $C_1$  and  $C_2$ ) are placed close and parallel to each other in such a way that distance between their centres is  $a$ . Find out the electric field intensity at a point  $P$  distant  $r$  from this system.  $\left( k = \frac{1}{4\pi\epsilon_0} \right)$

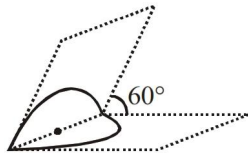


- (1) data insufficient (2)  $\frac{k\lambda_0 a}{r}$   
(3)  $\frac{k\lambda_0 a}{2r}$  (4)  $\frac{2k\lambda_0 a}{r}$
33. A non conducting infinite rod is placed along the z-axis : the upper half of the rod (lying along  $z \geq 0$ ) is charged positively with a uniform linear charge density  $+\lambda$  while the lower half ( $z < 0$ ) is charged negatively with a uniform linear charge density  $-\lambda$ . The origin is located at the junction of the positive and negative halves of the rod. A uniformly charged annular disc (surface charge density :  $\sigma_0$ ) of inner radius  $R$  and outer radius  $2R$  is placed in the x-y plane with its centre of the origin. The force on the rod due to the disc is

- (1)  $\frac{2\sigma_0\lambda R}{\epsilon_0}$  (2)  $\frac{\sigma_0\lambda R}{2\epsilon_0}$   
(3)  $\frac{\sigma_0\lambda R}{\epsilon_0}$  (4)  $\frac{\sigma_0\lambda R}{3\epsilon_0}$

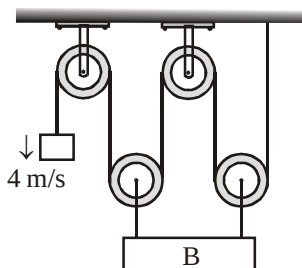
SPACE FOR ROUGH WORK

34. A uniformly charged ring of charge  $Q$  and radius  $R$  is fold from its diameter such that two halves make an angle  $60^\circ$  with each other then net electric field at centre of ring (O) is equal to



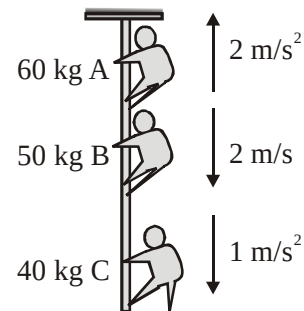
- (1)  $\frac{\sqrt{3}kQ}{\pi R^2}$  (2)  $\frac{2\sqrt{3}kQ}{\pi R^2}$   
(3)  $\frac{2kQ}{\pi R^2}$  (4)  $\frac{4kQ}{\pi R^2}$

35. For the given system find the velocity of block 'B' if velocity of block 'A' is 4 m/s in downward direction :- (Strings & pulley are ideal)



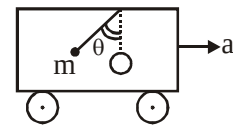
- (1)  $V_B = 1\text{ m/s} \uparrow$   
(2)  $V_B = 2\text{ m/s} \downarrow$   
(3)  $V_B = 4\text{ m/s} \uparrow$   
(4)  $V_B = \text{None}$

36. Tension in the rope at the rigid support is ( $g = 10\text{ m/s}^2$ ) :-



- (1) 760 N (2) 1360 N  
(3) 1580 N (4) 1620 N

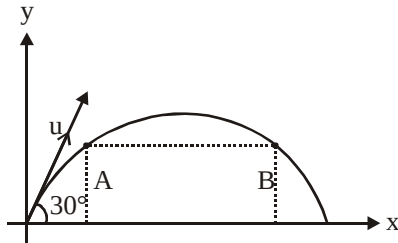
37. If trolley accelerates horizontally with acceleration  $a$  then bob is displaced backward from its initial vertical position. The angular deflection of the bob in equilibrium is :-



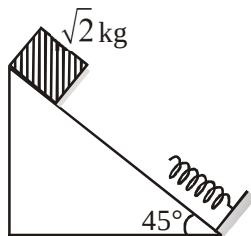
- (1)  $\theta = \cos^{-1} \left( \frac{a}{g} \right)$   
(2)  $\theta = \sin^{-1} \left( \frac{a}{g} \right)$   
(3)  $\theta = \cot^{-1} \left( \frac{a}{g} \right)$   
(4)  $\theta = \tan^{-1} \left( \frac{a}{g} \right)$

SPACE FOR ROUGH WORK

38. A particle is thrown with a velocity of  $v$  m/s. it passes A and B as shown in figure at time  $t_1 = 1$  s and  $t_2 = 3$  s. The value of  $u$  is ( $g = 10 \text{ m/s}^2$ ) :-

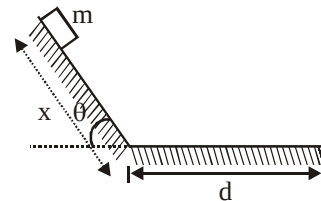


- (1) 20 m/s  
(2) 10 m/s  
(3) 40 m/s  
(4) 5 m/s
39. A block of mass  $\sqrt{2}$  kg is released from the top of an inclined smooth surface as shown in figure. If spring constant of spring is 100 N/m and block comes to rest after compressing the spring by 1m, then the distance travelled by block before it comes to rest is :-



- (1) 1 m  
(2) 1.25 m  
(3) 2.5 m  
(4) 5 m

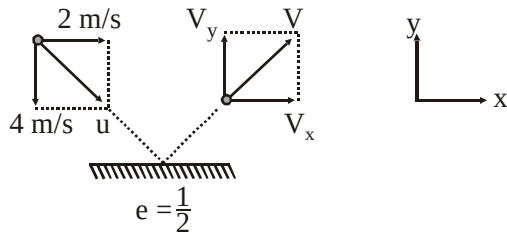
40. A block of mass  $m$  is released on the top of a smooth inclined plane of length  $x$  and inclination  $\theta$  as shown in figure. Horizontal surface is rough. If block comes to rest after moving a distance  $d$  on the horizontal surface, then coefficient of friction between block and surface is :-



- (1)  $\frac{x \sin \theta}{2d}$                       (2)  $\frac{x \cos \theta}{2d}$   
(3)  $\frac{x \sin \theta}{d}$                       (4)  $\frac{x \cos \theta}{d}$
41. A particle is projected from ground at an angle  $\theta$  with horizontal with speed  $u$ . The ratio of radius of curvature of its trajectory at point of projection to radius of curvature at maximum height is :-
- (1)  $\frac{1}{\sin^2 \theta \cos \theta}$   
(2)  $\cos^2 \theta$   
(3)  $\frac{1}{\sin^3 \theta}$   
(4)  $\frac{1}{\cos^3 \theta}$

SPACE FOR ROUGH WORK

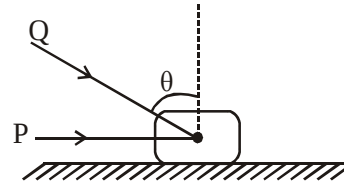
42. In the figure shown, a small ball hits obliquely a smooth and horizontal surface with speed  $u$  whose  $x$  and  $y$  components are indicated. If the coefficient of restitution is  $\frac{1}{2}$ , then its  $x$  and  $y$  components  $v_x$  and  $v_y$  just after collision are respectively :-



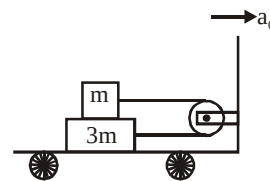
- (1) 4 m/s, 1 m/s      (2) 2 m/s, 1 m/s  
(3) 2 m/s, 2 m/s      (4) 4 m/s, 2 m/s
43. A smooth block is released at rest on a  $45^\circ$  incline and then slides a distance  $d$ . The time taken to slide is  $n$  times as much to slide on rough incline than on a smooth incline. The coefficient of friction is

- (1)  $\mu_k = 1 - \frac{1}{n^2}$   
(2)  $\mu_k = \sqrt{1 - \frac{1}{n^2}}$   
(3)  $\mu_s = 1 - \frac{1}{n^2}$   
(4)  $\mu_s = \sqrt{1 - \frac{1}{n^2}}$

44. A block of mass  $m$  lying on a rough horizontal plane is acted upon by a horizontal force  $P$  and another force  $Q$  inclined at an angle  $\theta$  to the vertical. The block will remain in equilibrium if the coefficient of friction between it and the surface is :-



- (1)  $\frac{P + Q \sin \theta}{mg + Q \cos \theta}$   
(2)  $\frac{P \cos \theta + Q}{mg - Q \sin \theta}$   
(3)  $\frac{P + Q \cos \theta}{mg + Q \sin \theta}$   
(4)  $\frac{P \sin \theta + Q}{mg - Q \cos \theta}$
45. In the given figure if all the surfaces are smooth then the acceleration of the two boxes relative to the trolley is.



- (1)  $3a_0/2$       (2)  $a_0$   
(3)  $a_0/2$       (4)  $2a_0$

SPACE FOR ROUGH WORK

46. The elements with atomic number 117 and 120 are yet to be discovered. In which group would you place these elements when discovered?  
(1) 17, 2 (2) 16, 4  
(3) 15, 3 (4) 18, 2
47. According to modern periodic law the properties of elements repeat at regular intervals when the elements are arranged in order of :-  
(1) decreasing atomic number  
(2) increasing atomic weight  
(3) increasing atomic number  
(4) decreasing atomic weight
48. In the fourth period of the periodic table, how many elements have one or more 4d electrons?  
(1) 2 (2) 18 (3) 0 (4) 6
49. Consider the following information about element P and Q :-
- |   | Period number | Group number |
|---|---------------|--------------|
| P | 2             | 15           |
| Q | 3             | 2            |
- Then formula of the compound formed by P and Q element is :-  
(1) PQ (2)  $P_3Q_2$   
(3)  $P_2Q_3$  (4)  $PQ_2$
50. Which of the following anion has the smallest radius?  
(1)  $H^-$  (2)  $F^-$  (3)  $Cl^-$  (4)  $Br^-$
51. The ionic radii of  $Li^+$ ,  $Be^{2+}$  and  $B^{3+}$  follow the order :  
(1)  $Be^{2+} > B^{3+} > Li^+$  (2)  $Li^+ > B^{3+} > Be^{2+}$   
(3)  $B^{3+} > Be^{2+} > Li^+$  (4)  $Li^+ > Be^{2+} > B^{3+}$
52. If the aufbau principle had not been followed, Ca ( $Z = 20$ ) would have been placed in the :-  
(1) s-block (2) p-block  
(3) d-block (4) f-block
53. The correct order of atomic size for group-13 is :-  
(1)  $B < Al < Ga < In$   
(2)  $B < Ga < In < Al$   
(3)  $B < Ga < Al < In$   
(4)  $B < In < Ga < Al$
54. The largest difference in atomic radii down the group in p-block elements takes place from :-  
(1) P-2 to P-3 (2) P-3 to P-4  
(3) P-4 to P-5 (4) P-5 to P-6
55. The correct order of second I.E. of C, N, O and F are in the order :-  
(1)  $F > O > N > C$  (2)  $C > N > O > F$   
(3)  $O > N > F > C$  (4)  $O > F > N > C$
56. Ionization energies of five elements in kcal/mol are given below :-
- | Atom | I   | II   | III  |
|------|-----|------|------|
| P    | 300 | 549  | 920  |
| Q    | 99  | 734  | 1100 |
| R    | 118 | 1091 | 1652 |
| S    | 176 | 347  | 1848 |
| T    | 497 | 947  | 1500 |
- The element having most stable oxidation state +2 is?  
(1) Q (2) R  
(3) S (4) T

SPACE FOR ROUGH WORK

57. Which of the following processes involves absorption of energy?
- $S_{(g)} + e^- \longrightarrow S_{(g)}^-$
  - $S_{(g)}^- + e^- \longrightarrow S_{(g)}^{2-}$
  - $Cl_{(g)} + e^- \longrightarrow Cl_{(g)}^-$
  - $Na_{(g)} + e^- \longrightarrow Na_{(g)}^-$
58. Second electron gain enthalpy :-
- is always negative
  - is always positive
  - can be positive or negative
  - is always zero
59. Which of the following represents correct order of electron affinity?
- $Cl > F > S > O$
  - $F > O > S > Cl$
  - $F > Cl > S > O$
  - $Cl > S > O > F$
60. The formation of the oxide ion  $O^{2-}(g)$  requires first an exothermic and then an endothermic step as shown below :-
- $$O(g) + e^- \longrightarrow O^-(g); \Delta H = -142 \text{ kJ mol}^{-1}$$
- $$O^-(g) + e^- \longrightarrow O^{2-}(g); \Delta H = 844 \text{ kJ mol}^{-1}$$
- This is because :-
- $O^-$  ion has comparatively larger size than oxygen atom
  - Oxygen has high electron affinity
  - $O^-$  ion will tend to resist the addition of another electron
  - Oxygen is more electronegative
61.  $3 N_0/2$  atoms of  $X_{(g)}$  are converted in to  $X_{(g)}^+$  by energy  $E_1$ ,  $2 N_0/3$  atoms of  $X_{(g)}$  are converted in  $X_{(g)}^-$  by energy  $E_2$ . Hence, ionisation potential and electron affinity of  $X_{(g)}$  are :- ( $N_0$  = Avogadro's number)
- $\frac{2E_1}{3N_0}, \frac{2E_2}{3N_0}$
  - $\frac{2E_1}{3N_0}, \frac{3E_2}{2N_0}$
  - $\frac{3E_1}{2N_0}, \frac{3E_2}{2N_0}$
  - $\frac{3E_1}{2N_0}, \frac{2E_2}{3N_0}$
62. Which of the following is different from other three oxides?
- MgO
  - SnO
  - ZnO
  - PbO
63. The X-X bond length is  $1.00 \text{ \AA}$  and C-C bond length is  $1.54 \text{ \AA}$ . If electronegativities of 'X' and 'C' are 3.0 and 2.0 respectively, the C-X bond length is likely to be :-
- $1.27 \text{ \AA}$
  - $1.18 \text{ \AA}$
  - $1.08 \text{ \AA}$
  - $1.28 \text{ \AA}$
64. Select CORRECT order of acidic strength of oxides :-
- $SiO_2 < P_4O_{10} < SO_3 < Cl_2O_7$
  - $Na_2O < MgO < SO_2 < SO_3$
  - $CO_2 < N_2O_5 < SO_3 < Cl_2O_7$
  - All are correct
65. The second ionisation enthalpies of Na, Mg, Al and Si are in the order:
- $Mg < Si < Al < Na$
  - $Na > Mg > Al > Si$
  - $Na < Mg < Al < Si$
  - $Na > Mg > Al < Si$

SPACE FOR ROUGH WORK

66. Be & Mg have +ve value of  $\Delta H_{eg}$  this can be explained by :-  
 (1) By their stable configuration  
 (2) By their extremely small size  
 (3) By weak shielding of 's' electrons  
 (4) By strong shielding of 's' electrons
67. A gas behaves like an ideal gas at  
 (1) high pressure and low temperature  
 (2) low pressure and high temperature  
 (3) high pressure and high temperature  
 (4) low pressure and low temperature
68. The density of nitrogen is maximum at  
 (1) STP  
 (2) 273 K and 2 atm  
 (3) 546 K and 1 atm  
 (4) 546 K and 2 atm
69. 1.0 litre of  $N_2$  and  $\frac{7}{8}$  litre of  $O_2$  at the same temperature and pressure were mixed together. What is the relation between the masses of the two gases in the mixture?  
 (1)  $M_{N_2} = 3M_{O_2}$  (2)  $M_{N_2} = 8M_{O_2}$   
 (3)  $M_{N_2} = M_{O_2}$  (4)  $M_{N_2} = 16M_{O_2}$
70. The density of a gas A is twice that of a gas B at the same temperature. The molecular weight of gas B is thrice that of A. The ratio of the pressures acting on A and B will be  
 (1) 1 : 4 (2) 7 : 8  
 (3) 2 : 5 (4) None of these
71. A volume  $V$  of a gas at a temperature  $T_1$  and a pressure  $p$  is enclosed in a sphere. It is connected to another sphere of volume  $V/2$  by a tube and stopcock. The second sphere is initially evacuated and the stopcock is closed. If the stopcock is opened the temperature of the gas in the second sphere becomes  $T_2$ . The first sphere is maintained at a temperature  $T_1$ . What is the final pressure  $p_1$  within the apparatus?  
 (1)  $\frac{2pT_2}{2T_2 + T_1}$  (2)  $\frac{2pT_2}{T_2 + 2T_1}$   
 (3)  $\frac{pT_2}{2T_2 + T_1}$  (4)  $\frac{2pT_2}{T_1 + T_2}$
72. Air contains 23% oxygen and 77% nitrogen by weight. The percentage of  $O_2$  by volume is  
 (1) 28.1 (2) 20.7  
 (3) 21.8 (4) 23.0
73. A mixture contains  $N_2O_4$  and  $NO_2$  in the ratio 2 : 1 by volume. The vapour density of the mixture is  
 (1) 45.4 (2) 49.8  
 (3) 32.6 (4) 38.3
74. An open vessel at  $27^\circ C$  is heated until  $3/8$ th of the air in it has been expelled. Assuming that the volume of vessel remains constant, calculate the temperature at which the vessel was heated.  
 (1)  $307^\circ C$  (2)  $107^\circ C$   
 (3)  $480^\circ C$  (4)  $207^\circ C$

SPACE FOR ROUGH WORK

75. A bottle of dry ammonia and one of dry hydrogen chloride are connected through a long tube. The stoppers at both ends of the tube are opened simultaneously. The white ammonium chloride ring first formed will be
- at the centre of the tube
  - near the hydrogen chloride bottle
  - near the ammonia bottle
  - throughout the length of the tube
76. At stp, 0.48 g of  $O_2$  diffused through a porous partition in 1200 seconds. What volume of  $CO_2$  will diffuse in the same time and under the same conditions?
- 286.5 mL
  - 346.7 mL
  - 112.2 mL
  - 224.8 mL
77. The molecular velocities of two gases at the same temperature are  $u_1$  and  $u_2$  and their masses are  $m_1$  and  $m_2$  respectively. Which of the following expressions are correct?
- $\frac{m_1}{u_1^2} = \frac{m_2}{u_2^2}$
  - $m_1 u_1 = m_2 u_2$
  - $\frac{m_1}{u_1} = \frac{m_2}{u_2}$
  - $m_1 u_1^2 = m_2 u_2^2$
78. The rms speed of  $N_2$  molecules in a gas is  $u$ . If the temperature is doubled and the nitrogen molecules dissociate into nitrogen atoms, the rms speed becomes
- $u/2$
  - $2u$
  - $4u$
  - $14u$
79. To which of the following gaseous mixtures is Dalton's law not applicable?
- $Ne + He + SO_2$
  - $NH_3 + HCl + HBr$
  - $O_2 + N_2 + CO_2$
  - $N_2 + H_2 + O_2$
80. The translational kinetic energy of an ideal gas depends only on its
- pressure
  - force
  - temperature
  - molar mass
81. A bubble of gas released at the bottom of a lake increases to eight times its original volume when it reaches the surface. Assuming that one atm pressure is equivalent to the pressure exerted by a column of water 10 m high, what is the depth of the lake?
- 90 m
  - 10 m
  - 70 m
  - 80 m
82. At what temperature is the rms speed of hydrogen molecules the same as that of oxygen molecules at  $1327^\circ C$ ?
- 173 K
  - 100 K
  - 400 K
  - 523 K
83. The compressibility of a gas is less than unity at stp. Therefore.
- $V_m > 22.4 \text{ L}$
  - $V_m < 22.4 \text{ L}$
  - $V_m = 22.4 \text{ L}$
  - $V_m = 44.8 \text{ L}$

SPACE FOR ROUGH WORK

84. At low pressure, the van der Waals equation is reduced to

(1)  $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RTV_m}$

(2)  $Z = \frac{pV_m}{RT} = 1 + \frac{b}{RT}p$

(3)  $pV_m = RT$

(4)  $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RT}$

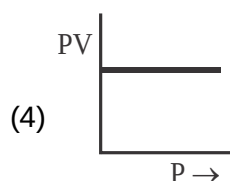
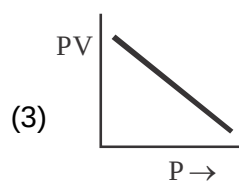
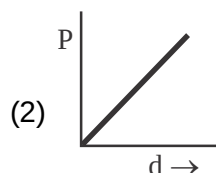
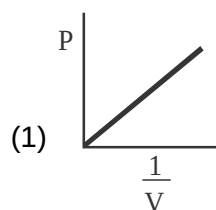
85. The temperature at which real gases obey the ideal gas laws over a wide range of pressures is called

- (1) critical temperature
- (2) inversion temperature
- (3) Boyle temperature
- (4) reduced temperature

86. For real gases compressibility factor is

- (1)  $z = 1$
- (2)  $z < 1$
- (3)  $z > 1$
- (4) Both (2) and (3)

87. Which is not a correct representation of Boyle's law?



88. For gaseous state, if most probable speed is denoted by  $C^*$ , average speed by  $\bar{C}$  and root mean square speed by  $C$ , then for a large number of molecules, the ratios of these speeds are

(1)  $C^* : \bar{C} : C = 1.225 : 1.128 : 1$

(2)  $C^* : \bar{C} : C = 1.228 : 1.225 : 1$

(3)  $C^* : \bar{C} : C = 1 : 1.128 : 1.225$

(4)  $C^* : \bar{C} : C = 1 : 1.225 : 1.128$

89. The ratio of van der Waals' constants  $a$

and  $b$ ,  $\left(\frac{a}{b}\right)$  has the dimension of

- (1)  $\text{atm L}^{-1}$
- (2)  $\text{L atm mol}^{-1}$
- (3)  $\text{L mol}^{-1}$
- (4)  $\text{atm L mol}^{-2}$

90. Match Column I with Column II and select the correct answers.

| Column I                 | Column II            |
|--------------------------|----------------------|
| A. Critical temperature  | 1. $\frac{a}{Rb}$    |
| B. Boyle's temperature   | 2. $\frac{2a}{Rb}$   |
| C. Inversion temperature | 3. $\frac{T}{T_c}$   |
| D. Reduced temperature   | 4. $\frac{8a}{27Rb}$ |

CODES

| A     | B | C | D |
|-------|---|---|---|
| (1) 2 | 1 | 4 | 3 |
| (2) 4 | 3 | 2 | 1 |
| (3) 2 | 3 | 4 | 1 |
| (4) 4 | 1 | 2 | 3 |

SPACE FOR ROUGH WORK

91. If the corpus callosum is removed in mammalian brain then what will be affected:-
- (1) Coordination of cerebrum
  - (2) Involuntary activity of brain
  - (3) Coordination of cerebellum
  - (4) Behaviour of cerebellum
92. Nerve impulse for hearing originates in-
- (1) Ear drum
  - (2) Auditory nerve
  - (3) Ear ossicles
  - (4) Cochlear nerve
93. Acetylcholine is:
- (1) Toxin
  - (2) Vitamin
  - (3) Enzyme
  - (4) Chemical transmitter
94. If both electrodes of cathode ray oscilloscope (CRO) are placed inside an axon, then it shows a membrane potential of \_\_\_\_ in resting state.
- (1) -70 mV
  - (2) -55 mV
  - (3) +30 mV
  - (4) 0 mV
95. Nervous band connecting the two cerebral hemispheres is:
- (1) Corpus albicans
  - (2) Corpus callosum
  - (3) Corpus striatum
  - (4) Corpus spongiosum
96. Subarachnoid space occurs
- (1) Above piamater
  - (2) Above durameter
  - (3) Inside blastocoele
  - (4) None of the above
97. Hypothalamus is:
- (1) Helpful in sleep
  - (2) Has centres for thirst and hunger
  - (3) Controls body temperature
  - (4) All the above
98. Third, sixth and eleventh cranial nerves are-
- (1) Oculomotor, Abducens and Accessory spinal
  - (2) Oculomotor, Trigeminal and Accessory spinal
  - (3) Optic, Facial and Accessory spinal
  - (4) Trochlear, Abducens and Vagus
99. What is the effect of parasympathetic nervous system
- (1) Originates heart beat
  - (2) Decrease heart beat
  - (3) Increases heart beat
  - (4) No effect on H.B.
100. Fifth cranial nerve is
- (1) Optic
  - (2) Trigeminal
  - (3) Abducens
  - (4) Facial
101. Brain ventricles are lined by-
- (1) Neurons
  - (2) Schwann cells
  - (3) Neuroglia
  - (4) Ependymal cells
102. Meninges form outer envelope in-
- (1) Cartilage
  - (2) Muscle
  - (3) Brain
  - (4) Kidney
103. Brain originates from-
- (1) Endoderm
  - (2) Mesoderm
  - (3) Ectoderm
  - (4) Ecto-endodermal
104. Hypophyseal portal system occurs in-
- (1) Brain
  - (2) Kidney
  - (3) Liver
  - (4) Heart
105. Nissl granules of neuron are made up of-
- (1) Proteins
  - (2) Mitochondria
  - (3) Ribonucleo protein
  - (4) DNA
106. The nerve secreting epinephrine or norepinephrine is called-
- (1) Adrenergic
  - (2) Cholinergic
  - (3) Ammonelic
  - (4) None of them

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p>107. Cerebrospinal fluid is formed by-</p> <p>(1) Nerve                      (2) Choroid plexus</p> <p>(3) Neurons                (4) Neuroglial cells</p> <p>108. Respiration, heart beat and peristalsis are controlled by-</p> <p>(1) Medulla oblongata</p> <p>(2) Medulla oblongata and cerebellum</p> <p>(3) Medulla oblongata and cerebrum</p> <p>(4) Cerebellum</p> <p>109. Which cells do not divide after birth-</p> <p>(1) Nerve cells              (2) Germ cells</p> <p>(3) Muscle cells            (4) Osteoblast</p> <p>110. Nerve impulse travels as-</p> <p>(1) Chemical impulse</p> <p>(2) Electrical impulse</p> <p>(3) Electrochemical impulse</p> <p>(4) Physical impulse</p> <p>111. Arbor vitae is composed of-</p> <p>(1) Grey matter            (2) White matter</p> <p>(3) Neuroglia cells      (4) All the above</p> <p>112. The most common synapse in our body is</p> <p>(1) Electrical</p> <p>(2) Chemical</p> <p>(3) Both are equally present</p> <p>(4) None of these</p> <p>113. Cell body (soma) of motor nerves is situated in-</p> <p>(1) Dorsal root of spinal cord</p> <p>(2) Dorsal horn of spinal cord</p> <p>(3) Ventral and lateral horn of spinal cord</p> <p>(4) Ventral root of spinal nerve</p> <p>114. Fourth cranial nerve is-</p> <p>(1) Trigeminal              (2) Abducens</p> <p>(3) Trochlear              (4) Optic</p> <p>115. Sympathetic nervous system increases-</p> <p>(1) Heart beat</p> <p>(2) Secretion of saliva</p> <p>(3) Secretion of digestive juices</p> <p>(4) All the above</p> <p>116. Main function of cerebellum is-</p> <p>(1) Balancing              (2) Sight</p> <p>(3) Hearing                  (4) Remembering</p> | <p>117. Centre for sense of smell is-</p> <p>(1) Cerebellum              (2) Olfactory lobes</p> <p>(3) Cerebrum              (4) Midbrain</p> <p>118. Nerve related to diaphragm is-</p> <p>(1) Vagus</p> <p>(2) Phrenic</p> <p>(3) Trigeminal</p> <p>(4) Glossopharyngeal</p> <p>119. Which of the following is the longest nerve-</p> <p>(1) Vagus</p> <p>(2) Olfactory</p> <p>(3) Hypoglossal</p> <p>(4) Glossopharyngeal</p> <p>120. In an axon, nerve impulse travels-</p> <p>(1) Away from cyton</p> <p>(2) Towards cyton</p> <p>(3) Both away and towards cyton</p> <p>(4) Not known</p> <p>121. The junction between axon of one neuron and dendrites of the next neuron is called-</p> <p>(1) junction point        (2) A joint</p> <p>(3) A synapse              (4) Constant</p> <p>122. The respiratory centre controlling inspiration and expiration are located in-</p> <p>(1) Medulla oblongata</p> <p>(2) Cerebellum</p> <p>(3) Diencephalon</p> <p>(4) Spinal cord</p> <p>123. Brain depends on blood for the supply of-</p> <p>(1) Oxygen and glucose</p> <p>(2) ATP and glucose</p> <p>(3) Oxygen and ATP</p> <p>(4) Oxygen and electrolytes</p> <p>124. In motor neuron the impulse following in which direction ?</p> <p>(1) In two direction</p> <p>(2) In one direction</p> <p>(3) In all direction</p> <p>(4) None of these</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

125. During development of action potential in a nerve fibres, positive and negative charges on outer and inner side of axon membrane are reversed due to-
- Extension of all  $K^+$  ions
  - More  $K^+$  ions enter than  $Na^+$  ions leave than axon
  - More  $Na^+$  ions enter the axon than  $K^+$  ions leaves the same
  - All  $Na^+$  ions enter the axon
126. Saltatory conduction of nerve impulse takes place through
- Dendrite
  - Myelinated axon
  - Non-myelinated axon
  - Cyton
127. Which one occurs in mesencephalon/midbrain-
- Cerebellum
  - Inferior colliculus
  - Thalamus
  - Mammillary body
128. Neurotransmitter between neuron and a muscle cell is-
- Dopamine
  - Serotonin
  - GABA
  - Acetylcholine
129. Medulla oblongata encloses-
- Fourth ventricle
  - Third ventricle
  - Second ventricle
  - Optic lobes
130. At resting stage nerve cells has-
- Low  $K^+$  outside and high  $Na^+$  inside
  - High  $K^+$  inside and high  $Na^+$  outside
  - High  $K^+$  inside and low  $Na^+$  outside
  - High  $K^+$  outside and low  $Na^+$  inside
131. In a human being number of cranial nerves are-
- 10 pairs
  - 20 pairs
  - 6 pairs
  - 12 pairs
132. Depolarization of nerve impulse involves-
- Entry of  $Na^+$  only
  - Entry of  $K^+$  only
  - Entry of  $Na^+$  and exit of  $K^+$
  - Entry of  $K^+$  and exit of  $Na^+$
133. Corpus callosum is found in the brain of-
- Pigeon
  - Crocodile
  - Frog
  - Elephant
134. Node of Ranvier is found in-
- Muscle bundles
  - Dendrite
  - Axon
  - Right auricle
135. A thin vascular membrane of brain is-
- Arachnoid
  - Piameter
  - Durameter
  - Epineurium
136. The threshold stimulus for most excitable cells is
- +30 mV
  - +70 mV
  - 70 mV
  - 55 to -60 mV
137. Which of the following nerve innervates upper jaw?
- Trigeminal
  - Pathetic
  - Vagus
  - Oculomotor
138. Influx of sodium ions from extracellular fluid to inside the nerve cell causes
- Repolarisation
  - Depolarisation
  - Polarisation
  - Hyperpolarisation
139. The main cause of paralysis is-
- Some defect in muscles
  - Complete destruction of sensory nerve
  - Complete destruction of motor nerve
  - None of the above
140. Most neurons in cerebral cortex of humans are
- Unipolar
  - Bipolar
  - Apolar
  - Multipolar
141. Mark the odd one w.r.t. origin.
- Astrocyte
  - Microgliocytes
  - Neuron
  - Oligodendrocyte
142. The part of the brain which is most developed in higher primates is :
- cerebellum
  - cerebrum
  - medulla oblongata
  - olfactory lobes

143. The cerebellum is concerned with :
- (1) memory
  - (2) perception
  - (3) intelligence
  - (4) coordination of muscular movement
144. A mass of cytons is called as :
- (1) synapse                      (2) ganglion
  - (3) motor nerve                (4) nerve fibre
145. The pathway of reflex arc as :
- (1) sense organ, spinal cord, motor nerve, sensory nerve, muscle
  - (2) sense organ, sensory nerve, spinal cord, motor nerve, muscle
  - (3) sense organ, motor nerve, spinal cord, sensory nerve, muscle
  - (4) sense organ, motor nerve, spinal cord, sensory nerve, muscle
146. The sense of balance maintained with the help of
- (1) hammer
  - (2) stirrup
  - (3) semicircular canals
  - (4) cochlea
147. Which of the following is mismatched-
- (1) Cerebrum-Memory
  - (2) Cerebellum-Equilibrium of body
  - (3) Olfactory lobes - Smell
  - (4) Medulla - Temperature
148. Which of the following nerve supplies organ of corti ?
- (1) Auditory                      (2) Olfactory
  - (3) Trochlear                    (4) Vagus
149. Which is not a part of hindbrain-
- (1) Thalamus                      (2) Cerebellum
  - (3) Pons                            (4) Medulla
150. Corti's organ is present in-
- (1) Reisner's membrane
  - (2) Scala vestibuli
  - (3) Basilar's membrane
  - (4) Middle lamella
151. Parkinson's disease (characterized by tremors and progressive rigidity of limbs) is caused by degeneration of brain neurons that involved in movement control and make use of neurotransmitter-
- (1) Acetylcholine                (2) Norepinephrine
  - (3) Dopamine                    (4) GABA
152. In a man abducens nerve is injured. Which one of the following functions will be affected-
- (1) Movement of the eye ball
  - (2) Movement of the tongue
  - (3) Swallowing
  - (4) Movement of the neck
153. Which of the following is the main factor responsible for resting membrane potential?
- (1) Presence of more sodium ions inside neuron
  - (2) Presence of negatively charged proteins outside neuron
  - (3) Presence of positively charged proteins inside the neuron
  - (4) More permeability of membrane for potassium ions in resting state
154. Which regulates the amount of light falling on the lens
- (1) Cornea                            (2) Retina
  - (3) Pupil                            (4) Ciliary body
155. Which of the following are the cavities of the inner ear ?
- (1) Outer bony Labyrinth
  - (2) Viterous cavity
  - (3) Inner membranous Labyrinth
  - (4) (1) and (3)
156. In glaucoma-
- (1) Eyeball elongates
  - (2) Eyeball shortened
  - (3) Lens becomes opaque
  - (4) fluid pressure increases in eye

157. Outer bony labyrinth is filled with a fluid called ?  
 (1) Endolymph  
 (2) Vitreous humour  
 (3) Perilymph  
 (4) Aqueous humour
158. Which is the correct sequence of the parts of the eye that the light rays cross in reaching the retina-  
 (1) Pupil → cornea → aqueous humour → lens → vitreous humour  
 (2) Lens → cornea → aqueous humour → vitreous humour → pupil  
 (3) Aqueous humour → vitreous humour → cornea → pupil → lens  
 (4) Cornea → aqueous humour → pupil → lens → vitreous humour
159. The space between the lens and the cornea of the human eye is-  
 (1) Vitreous chamber  
 (2) Aqueous chamber  
 (3) Retina  
 (4) Iris
160. Malleus is present in the-  
 (1) Inner ear (2) Outer ear  
 (3) Middle ear (4) Eye
161. The movement of eye ball is brought about by the-  
 (1) Adductor muscle (2) Rectus muscle  
 (3) Biceps (4) Triceps
162. The cavity of the eye is divided into aqueous chamber and vitreous chamber by-  
 (1) Optic nerve (2) Retina  
 (3) Lens (4) Iris
163. Semicircular canals occur in-  
 (1) Heart (2) Kidney  
 (3) Intestine (4) Ear
164. Middle ear has-  
 (1) Two sets of tiny bones  
 (2) A set of three tiny bones  
 (3) Two sets of large bones  
 (4) Three sets of large bones
165. External ears are characteristic of-  
 (1) Mammals (2) Reptiles  
 (3) Amphibians (4) Pisces
166. Scala tympani is connected with scala vestibuli by means by-  
 (1) Stapes  
 (2) Helicotrema  
 (3) Fenestra ovalis  
 (4) Fenestra rotundus
167. The defective condition of accommodation of the eye in which distant objects are seen distinctly but near objects are indistinct is-  
 (1) Astigmatism (2) Presbiopia  
 (3) Myopia (4) Hypermetropia
168. The iris of the eye is an extension of-  
 (1) Cornea (2) Sclerotic  
 (3) Retina (4) Choroid
169. In the chemistry of vision in mammals, the photosensitive substance is called  
 (1) Sclerotin (2) Retinol  
 (3) Melanin (4) Rhodopsin
170. Glaucoma is caused by-  
 (1) Opacity of lens  
 (2) Increase in the size of eyeball  
 (3) blockage of canal of Schlemm  
 (4) Drying of vitreous humor
171. Vitamin A combines with a protein in the retina to produce-  
 (1) Glaucoma  
 (2) Night blindness  
 (3) Rhodopsin  
 (4) Colour blindness
172. In human eye, at the blindspot-  
 (1) Only rods are present  
 (2) Only cones are present  
 (3) Both rods and cones are present  
 (4) Neither rods nor cones are present
173. Which one of the following is not a part of ear  
 (1) Eustachian tube  
 (2) Cone cell  
 (3) Utriculus  
 (4) Sacculus

174. The black pigment in the eye, which reduces internal reflection, is located in-

  - (1) Iris
  - (2) Sclerotic
  - (3) Choroid
  - (4) Cornea

175. Eustachian tube connects-

  - (1) Middle ear to nasopharynx
  - (2) Middle ear to external ear
  - (3) Left ventricle to right ventricle
  - (4) Left atrium to right atrium

176. Area of retina in human eye containing only cone cells is called-

  - (1) Macula lutea
  - (2) Fovea centralis
  - (3) Pecten
  - (4) Tapetum

177. The tympanic membrane is connected to?

  - (1) Stapes
  - (2) Incus
  - (3) Malleus
  - (4) All of the above

178. When we migrate from dark to light, we fail to see for sometimes but after a time visibility becomes normal. It is an example of-

  - (1) Mutation
  - (2) Adaptation
  - (3) Photoperiodism
  - (4) Accommodation

179. The place in eye of mammals from which optic nerves and blood vessels leaves the eye ball is-

  - (1) blind spot
  - (2) yellow spot
  - (3) pars optica
  - (4) none of these

180. Internal ear is filled with :

  - (1) perilymph
  - (2) endolymph
  - (3) lymph
  - (4) both (1) and (2)





# Result : NEET (UG) 2019

# ALLEN है... तो मुमकिन है

**AIR**

✓ **8 in Top 10 All India Ranks**

✓ **61 in Top 100 All India Ranks**

TOTAL STUDENTS QUALIFIED

**Classroom**  
**63415**

**95102\***

**Distance**  
**31687**

**1**

**NALIN KHANDELWAL**  
Classroom

|                                                                                                                                               |                                                                                                                                            |                                                                                                                                             |                                                                                                                                           |                                                                                                                                            |                                                                                                                                           |                                                                                                                                             |                                                                                                                                           |                                                                                                                                           |                                                                                                                                           |                                                                                                                                              |                                                                                                                                             |                                                                                                                                               |                                                                                                                                                |                                                                                                                                              |                                                                                                                                                |
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| <b>AIR 2</b><br><br><b>BHAVIK BANSAL</b><br>Distance       | <b>AIR 4</b><br><br><b>SWASTIK BHATIA</b><br>Distance   | <b>AIR 5</b><br><br><b>ANANT JAIN</b><br>Classroom       | <b>AIR 6</b><br><br><b>SARTHAK R. BHAT</b><br>Distance | <b>AIR 8</b><br><br><b>DHRUV KUSHWAHA</b><br>Distance  | <b>AIR 9</b><br><br><b>MIHIR RAI</b><br>Distance     | <b>AIR 10</b><br><br><b>RAGHAV DUBEY</b><br>Classroom  |                                                                                                                                           |                                                                                                                                           |                                                                                                                                           |                                                                                                                                              |                                                                                                                                             |                                                                                                                                               |                                                                                                                                                |                                                                                                                                              |                                                                                                                                                |
| <b>AIR-11</b><br><br><b>EMANT KHANDELIA</b><br>Distance    | <b>AIR-17</b><br><br><b>RICHAM AGARWAL</b><br>Distance  | <b>AIR-18</b><br><br><b>HARSHVI N. J.</b><br>Classroom   | <b>AIR-19</b><br><br><b>ARUNANGSHU B.</b><br>Classroom | <b>AIR-20</b><br><br><b>SATVIK LODHA</b><br>Classroom   | <b>AIR-21</b><br><br><b>HARSH JAIN</b><br>Distance     | <b>AIR-22</b><br><br><b>HARDIK GUPTA</b><br>Classroom    | <b>AIR-24</b><br><br><b>LALIT AGARWAL</b><br>Classroom | <b>AIR-26</b><br><br><b>APOORV RAGHU</b><br>Classroom  | <b>AIR-27</b><br><br><b>PARTH GILL</b><br>Distance     | <b>AIR-28</b><br><br><b>GAUTAM CHELLANI</b><br>Distance  | <b>AIR-30</b><br><br><b>HARSH AGARWAL</b><br>Classroom | <b>AIR-37</b><br><br><b>LOHITASHYA GOYAL</b><br>Distance | <b>AIR-38</b><br><br><b>VISHWA RAKESH</b><br>Distance     | <b>AIR-39</b><br><br><b>ANURAG GAUTAM</b><br>Distance   | <b>AIR-44</b><br><br><b>M. H. SAYAN SHAH</b><br>Classroom |
| <b>AIR-45</b><br><br><b>SHIKHAR GUPTA</b><br>Distance      | <b>AIR-46</b><br><br><b>DIPRASOM DAS</b><br>Distance    | <b>AIR-47</b><br><br><b>BAHADUR SINGH</b><br>Distance    | <b>AIR-48</b><br><br><b>AMAN AGRAWAL</b><br>Classroom  | <b>AIR-49</b><br><br><b>VARUN CHOPRA</b><br>Distance    | <b>AIR-51</b><br><br><b>ISHAAN MISHRA</b><br>Distance  | <b>AIR-52</b><br><br><b>DISHA S. AGRAWAL</b><br>Distance | <b>AIR-53</b><br><br><b>SARVAGYA JAIN</b><br>Distance  | <b>AIR-54</b><br><br><b>HARRY GOYAL</b><br>Distance    | <b>AIR-56</b><br><br><b>ISHIKA GUPTA</b><br>Distance   | <b>AIR-58</b><br><br><b>SHREYAS AGRAWAL</b><br>Classroom | <b>AIR-60</b><br><br><b>ABHAY MITTAL</b><br>Classroom  | <b>AIR-61</b><br><br><b>CHIRAG JORVEKAR</b><br>Distance  | <b>AIR-63</b><br><br><b>VINEET HARSH</b><br>Distance      | <b>AIR-65</b><br><br><b>VANSHI K. NEELA</b><br>Distance | <b>AIR-66</b><br><br><b>SHREY MUDGAL</b><br>Classroom     |
| <b>AIR-69</b><br><br><b>PRIYANSHU AGARWAL</b><br>Classroom | <b>AIR-70</b><br><br><b>ISHIKA KAMRA</b><br>Classroom   | <b>AIR-71</b><br><br><b>STUTI KHANDWALA</b><br>Classroom | <b>AIR-72</b><br><br><b>NAVEEN SHARMA</b><br>Distance  | <b>AIR-75</b><br><br><b>SARTHAK SHARMA</b><br>Classroom | <b>AIR-76</b><br><br><b>VANSHIKA ARORA</b><br>Distance | <b>AIR-79</b><br><br><b>SAYAM V. BHATT</b><br>Classroom  | <b>AIR-81</b><br><br><b>UDITH MOHAN</b><br>Distance    | <b>AIR-84</b><br><br><b>PRAJNAPAN BASU</b><br>Distance | <b>AIR-85</b><br><br><b>HARSH RAMOLIYA</b><br>Distance | <b>AIR-86</b><br><br><b>UTKARSH SHARMA</b><br>Classroom  | <b>AIR-87</b><br><br><b>NISHTHA</b><br>Distance        | <b>AIR-89</b><br><br><b>ARYAN BANSAL</b><br>Distance     | <b>AIR-91</b><br><br><b>SANGEETA AGRAWAL</b><br>Classroom | <b>AIR-92</b><br><br><b>YASHDEEP</b><br>Classroom       | <b>AIR-93</b><br><br><b>ISHAN GUPTA</b><br>Classroom      |
| <b>AIR-94</b><br><br><b>MEGHA GUPTA</b><br>Classroom       | <b>AIR-96</b><br><br><b>GAUTAM KAUSHAL</b><br>Classroom | <b>AIR-97</b><br><br><b>POOJA YADAV</b><br>Distance      |                                                        |                                                                                                                                            |                                                                                                                                           |                                                                                                                                             |                                                                                                                                           |                                                                                                                                           |                                                                                                                                           |                                                                                                                                              |                                                                                                                                             |                                                                                                                                               | <b>AIR-99</b><br><br><b>PRAGYA MITRA</b><br>Distance      | <b>AIR-100</b><br><br><b>VIVEK KUMAR</b><br>Classroom   |                                                                                                                                                |

\*As Per Result Compiled so far

Authenticity of Result :  
Power of **ALLEN**



Heartiest  
Congratulations!